

sol 1 H. 20

Sort the following array using Quick sort

54	12	19	3	8	46	31
0	1	2	3	4	5	6

$n = 7$, first = 0, last = 6

i	j	a[i]	a[j]	a[pivot]
0	6	54	31	54

first qs call
 $qs(a, 0, 6)$

54	12	19	3	8	46	31
i	i	i	i	i	i	(2)
						(1)

$\Rightarrow i = j$

so, $a[j] \leftrightarrow a[pivot]$

$\Rightarrow 31 \leftrightarrow 54$

$\Rightarrow$

31	12	19	3	8	46	54
0	1	2	3	4	5	6

first = 0, last = 5

$\Rightarrow qs(a, 0, 5)$

i	j	a[i]	a[j]	a[pivot]
0	5	31	12	31

31	12	19	3	8	46
i	i	i	i	i	j
				(1)	(1)

$\Rightarrow i > j$

$\Rightarrow a[j] \leftrightarrow a[pivot]$

$\Rightarrow 8 \leftrightarrow 31$

$\Rightarrow$

8	12	19	3	31	46	54
0	1	2	3			

first = 0, last = 3 $\Rightarrow qs(a, 0, 3)$

i	j	a[i]	a[j]	a[pivot]
0	3	8	3	8

8	12	19	3
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i

(1)

(7)

$\Rightarrow i < j$

$\Rightarrow a[i] \leftrightarrow a[j]$

$\Rightarrow$ swap 12 & 3

$\Rightarrow$

8	3	19	12
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i

j

j

(7)

(2)

$\Rightarrow i > j$

$\Rightarrow a[j] \leftrightarrow a[\text{pivot}]$

$\Rightarrow$ swap 3 & 8

$\Rightarrow$

3	8	19	12	31	46	54
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0

1

2

3

4

5

6

first = 1

last = 3

$\Rightarrow \text{qs}(a, 1, 3)$

i	j	$a[i]$	$a[j]$	$a[\text{pivot}]$
1	3	8	12	8

8	19	12
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i

j

j

(7)

(2)

$\Rightarrow i > j$

$\Rightarrow a[j] \leftrightarrow a[\text{pivot}]$

$\Rightarrow$ swap 8 & 8

$\Rightarrow$

3	8	19	12	31	46	54
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0

1

2

3

4

5

6

first = 2

last = 3

$\Rightarrow \text{qs}(a, 2, 3)$

i	j	$a[i]$	$a[j]$	$a[\text{pivot}]$
2	3	19	12	19

19	12
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i

i

j

$i = j$

$\Rightarrow a[j] \leftrightarrow a[\text{pivot}]$

$\Rightarrow$ swap 12 & 19

$\therefore$ final sorted array :-

$\Rightarrow$

3	8	12	19	31	46	54
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